

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: STATISTICS 1 (3.1)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Mr Otieno, the principal of Lakeview Senior School in Kisumu, has money for only one option: a school bus route, a bicycle shed or a second lunch serving. In this Final Explanation you will use everything you learned about Grade 10 Green's travel data to answer the driving question: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision? Write your answers in the space under each prompt. Use numbers from the data to back up every claim, give units where needed (learners, seconds), and write the mean as 'mean'. You may use a calculator and a protractor if you need them. Remember that all the data come from one class only.

Part 1: From a messy list to a frequency table

In Lesson 1, three learners disagreed. Atieno said 'Almost everyone walks', Brian said 'Most of us come by matatu', and Zawadi said 'The bus is always full'. Grade 10 Green collected 40 responses in learner order, but the raw list did not settle the argument.

Here is the frequency table of the same 40 responses:

Travel mode	Frequency
Walk	14
Matatu	12
Boda boda	4
Bicycle	3
Private car	3

(a) The raw list is unorganised: the responses are in learner order, so you would have to count every W, M, B, Bi, C and S by eye to compare the groups. The frequency table groups the same 40 responses and gives one number for each travel mode, and the frequencies add to 40 ($14 + 12 + 4 + 3 + 3 + 4 = 40$), so the groups can be compared quickly.

(b) Percentage = $f / n \times 100$. Walk = $14/40 \times 100 = 35\%$. Matatu = $12/40 \times 100 = 30\%$. Together: 26 of $40 = 65\%$.

(c) Neither group is more than half (more than 20 learners). Walk is 14 of 40 and Matatu is 12 of 40. So 'Almost everyone walks' and 'Most of us come by matatu' are both wrong, although together they make 65% of the class. Zawadi's claim is about the school's single 14-seater bus serving the whole school. Only 4 of 40 learners (10%) in this class use the bus, so the class count can neither confirm nor refute her claim.

School bus	4
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(a) Explain why the raw list of 40 responses could not settle the claims, but the frequency table can. (b) Work out the percentage for Walk and for Matatu, and the percentage for the two groups together. (c) Use your numbers to decide whether Atieno and Brian are right, and explain what the class data can and cannot tell us about Zawadi's claim.

Part 2: Choosing a typical value

Grade 10 Green also counted the number of siblings each learner has:

Siblings	0	1	2	3	4	5
Learners	5	9	12	8	4	2

An Eldoret club group of 7

(a) Total learners = 40. Sum of $f \times x = 0 + 9 + 24 + 24 + 16 + 10 = 83$. Mean = $83 \div 40 = 2.075$, about 2.1 siblings. Mode = 2 siblings (highest frequency, 12). Median: cumulative frequencies are 5, 14, 26, so the 20th and 21st values are both 2, and the median is 2 siblings.

(b) Sum = 765. Mean = $765 \div 7 = 109.3$ s. Median = 4th value = 106 s. Mode = 105 s.

(c) The siblings data is spread fairly evenly, so the three measures are close (about 2). The runner times have one outlier, 130 s, which pulls the mean up to 109.3 s, higher than six of the seven times. Without it the mean of the other six is $635 \div 6 = 105.8$ s, while the median hardly changes. So the median (106 s) is the better measure for the runner times because it is not pulled by the outlier.

(d) Travel mode is categories, not numbers, so the mean and median cannot be found. Only the mode can be used: Walk (14 of 40). Note that Walk is the mode but is not a majority.

runners had these 800 m training times in seconds: 104, 105, 105, 106, 107, 108, 130 (the runner with 130 s had a cold). (a) Find the mean, median and mode of the siblings data. (b) Find the mean, median and mode of the runner times, giving the mean to 1 decimal place. (c) Why do the three measures agree for siblings but differ for the runner times? Which measure would you use for the runner times, and why? (d) Which measure can you use for the travel mode data, and what is its value?	
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Part 3: Choosing and checking graphs	
Mr Kamau showed two bar graphs of the same travel data. On one, the frequency axis started at 0. On the other, it started at 2. (a) Which type of graph is best for the travel data, and which is best for Mama Achieng's mandazi sales at Kibuye market (Mon 40, Tue 45, Wed 42, Thu 50, Fri 65, Sat 30, Sun 20)? Give a reason for each. (b) On the graph where the axis starts at 2, the Walk bar is 12 units tall and the Bicycle bar is 1 unit tall. What does it make walking look like compared with cycling? What is the true comparison from the data? (c) List four things you would check before trusting a graph.	(a) A bar graph is best for the travel data because travel mode is categories, so the bars are separate with equal widths and gaps. A line graph is best for the mandazi sales because they change over time, with days on the horizontal axis and points joined by straight lines. The mandazi line graph peaks on Friday (65) and is lowest on Sunday (20). (b) With the axis starting at 2, the Walk bar (12 units) looks 12 times the Bicycle bar (1 unit). The true comparison is 14 : 3, so walking is only about 4.7 times as many as cycling. The graph changes how the data looks, not the data itself. (c) Check that: the frequency axis starts at 0; the scale has equal steps; both axes are labelled and there is a title; for a pie chart, the angles add up to 360 degrees. Also read the numbers on the graph instead of trusting the picture alone.

Part 4: Pie chart angles

You drew a pie chart of the 40 travel responses.

(a) How many degrees does each learner represent? Show your working. (b) Calculate the angle for Walk and for Matatu. (c) Give one advantage of the pie chart and one advantage of the bar graph for the travel data. (d) What does the pie chart show about the bicycle group, and what does this mean for the bicycle shed?

(a) $360 \div 40 = 9$ degrees per learner.

(b) Angle = $f / n \times 360$. Walk = $14/40 \times 360 = 126$ degrees. Matatu = $12/40 \times 360 = 108$ degrees.

(c) The pie chart shows each group as a share of the whole, so you can see at a glance that neither Walk nor Matatu takes more than half of the circle. The bar graph shows exact counts and is better for comparing close values such as Boda boda 4 and School bus 4.

(d) The Bicycle sector is small: 3 of 40 learners (7.5%), which is 27 degrees. So this class gives little support for building a bicycle shed.

Part 5: Your Final Explanation for Mr Otieno

Write a short report (about 8 to 10 sentences) for Mr Otieno that answers the driving question: How can we turn a messy list of data from our school into clear evidence that helps our principal make a fair decision?

In your report: (1) describe the steps that turn the raw list into evidence; (2) say what the evidence shows about the three claims; (3) give your advice about the bicycle shed, the school bus route and the second lunch serving; (4) explain one limit of the data and how Mr Otieno should deal with it.

Dear Mr Otieno, the raw list of 40 responses from Grade 10 Green could not settle the argument, because it was unorganised. We tallied it into a frequency table (Walk 14, Matatu 12, Boda boda 4, Bicycle 3, Private car 3, School bus 4; total 40) and showed it in a bar graph and a pie chart. The mode of travel is Walk. Walk is 35% and Matatu is 30%. Neither is more than half, so 'almost everyone walks' and 'most of us come by matatu' are both wrong, although together they make 65%. Only 3 of 40 learners (7.5%) cycle, so this class gives little support for a bicycle shed, and we advise no money for a shed on this evidence. Only 4 of 40 (10%) use the school bus, but Zawadi's claim is about the single 14-seater bus serving the whole school, so our class cannot confirm or refute it. The data come from only one class, which is a small sample. We advise collecting the same data from the whole school before you decide between the bus route and the second lunch serving. The best summary or graph depends on the type of data and the question, so check that any graph has an axis starting at 0 before trusting it.

RUBRIC

Criterion

Excellent (4)

Proficient (3)

Developing (1–2)

Organising data and calculating from tables	All percentages, angles, means and medians are correct, with clear working and the frequencies checked against the total.	Most calculations are correct with some working shown; one small slip in rounding or working.	Several calculations are incorrect or have no working; totals are not checked.
Choosing measures and graphs	Chooses mean, median, mode and graph types correctly for each data type, and explains using the outlier and the categorical data.	Chooses most measures and graphs correctly with partial reasons.	Choices are unclear or not linked to the type of data.
Evidence-based reasoning about the claims	Uses exact numbers (14 of 40, 35%; 12 of 40, 30%; 65%) to judge each claim, and correctly says the class count cannot test Zawadi's bus claim.	Judges the claims with some numbers; the bus claim is mentioned but not fully explained.	Gives opinions without numbers, or states the claims are right or wrong without evidence.
Critical reading of graphs	Explains the truncated axis effect (12 times against a true ratio of about 4.7) and lists several correct checks for an honest graph.	Explains that the axis can mislead and lists some checks.	Does not recognise how a graph can mislead.
Final recommendation and communication	Clear, honest report that links the steps from raw list to evidence, advises no shed, recommends whole-school data, and states the limit of one class.	Gives a clear recommendation but misses one step or the limit of the sample.	Recommendation is unclear or is not supported by the data.

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